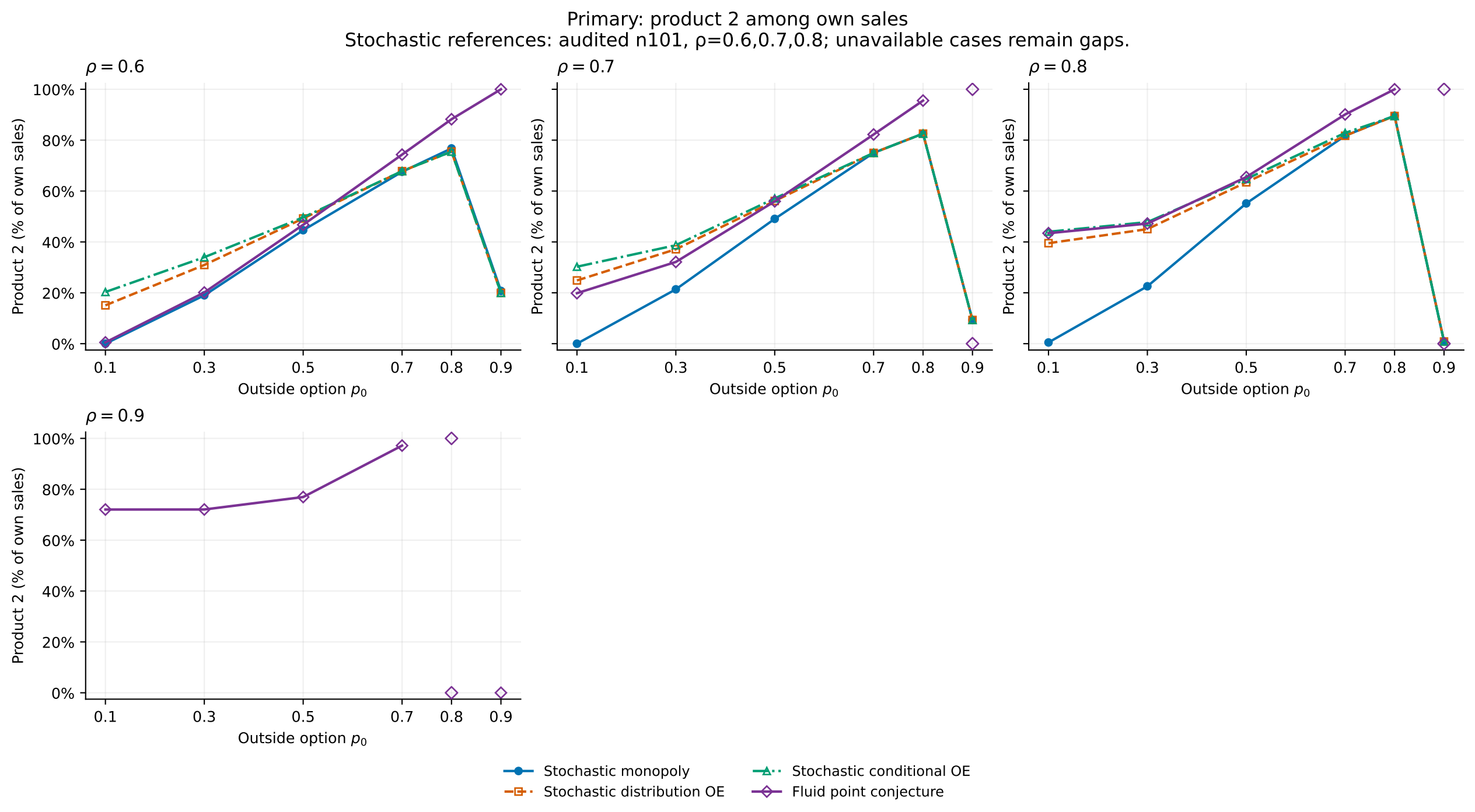
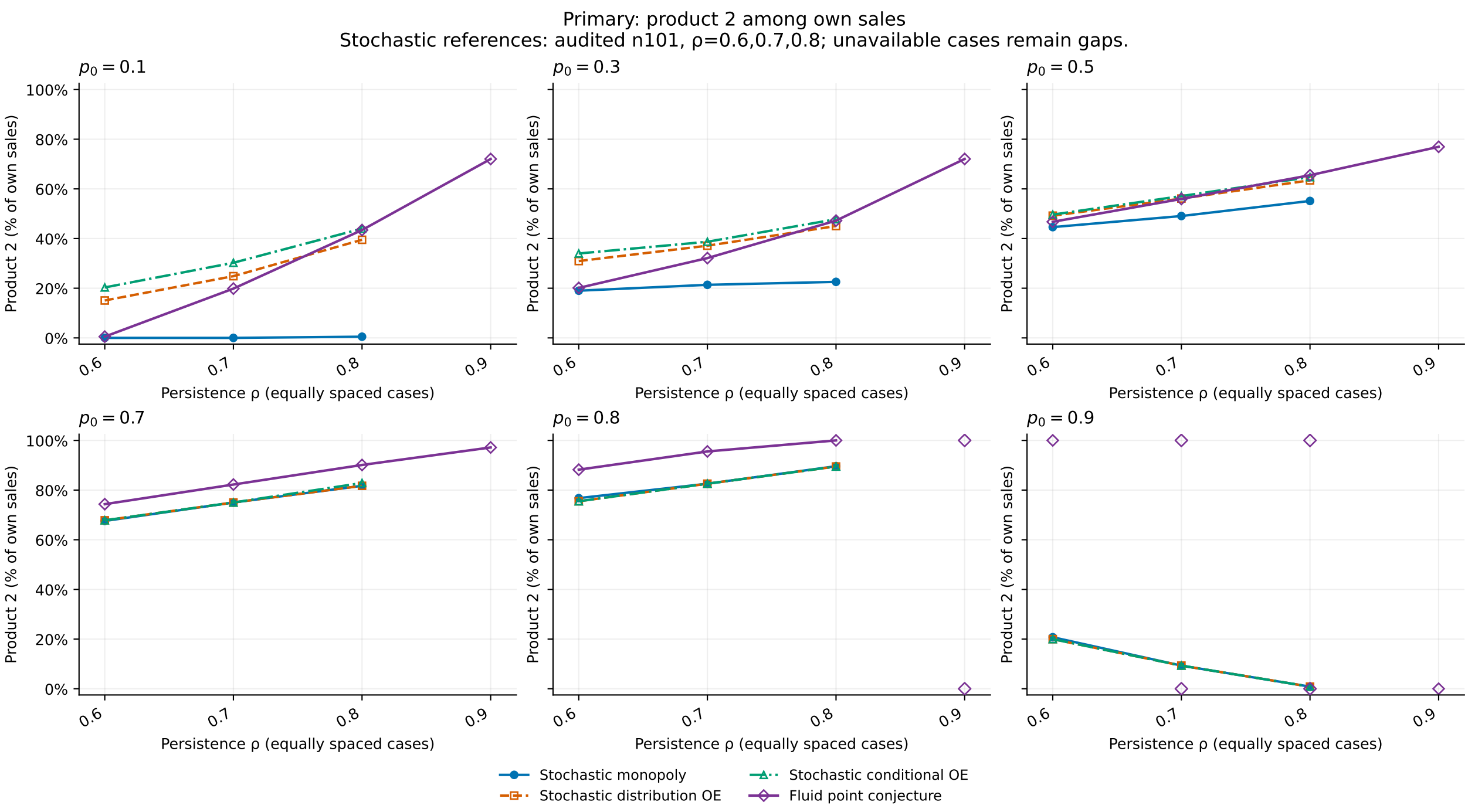
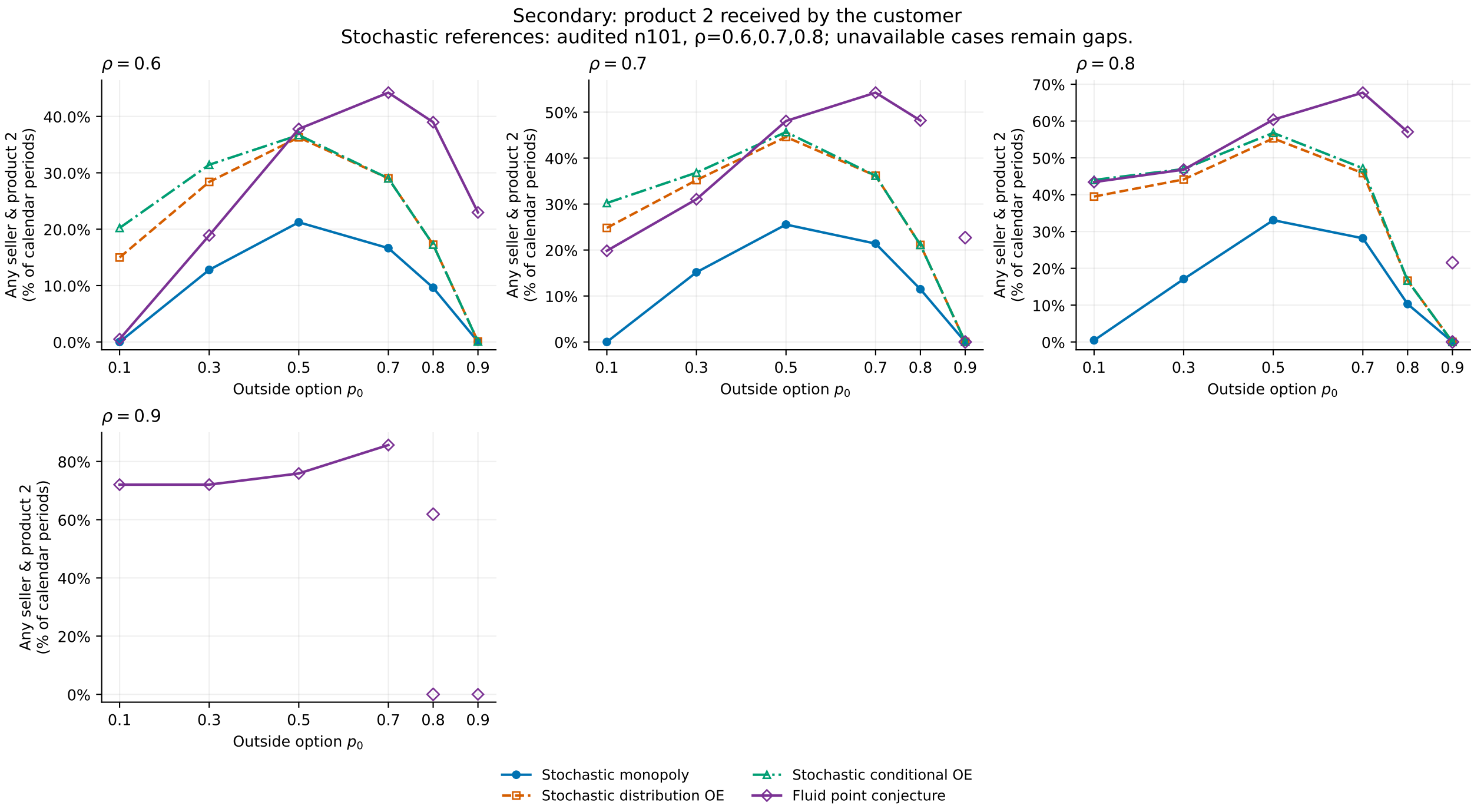
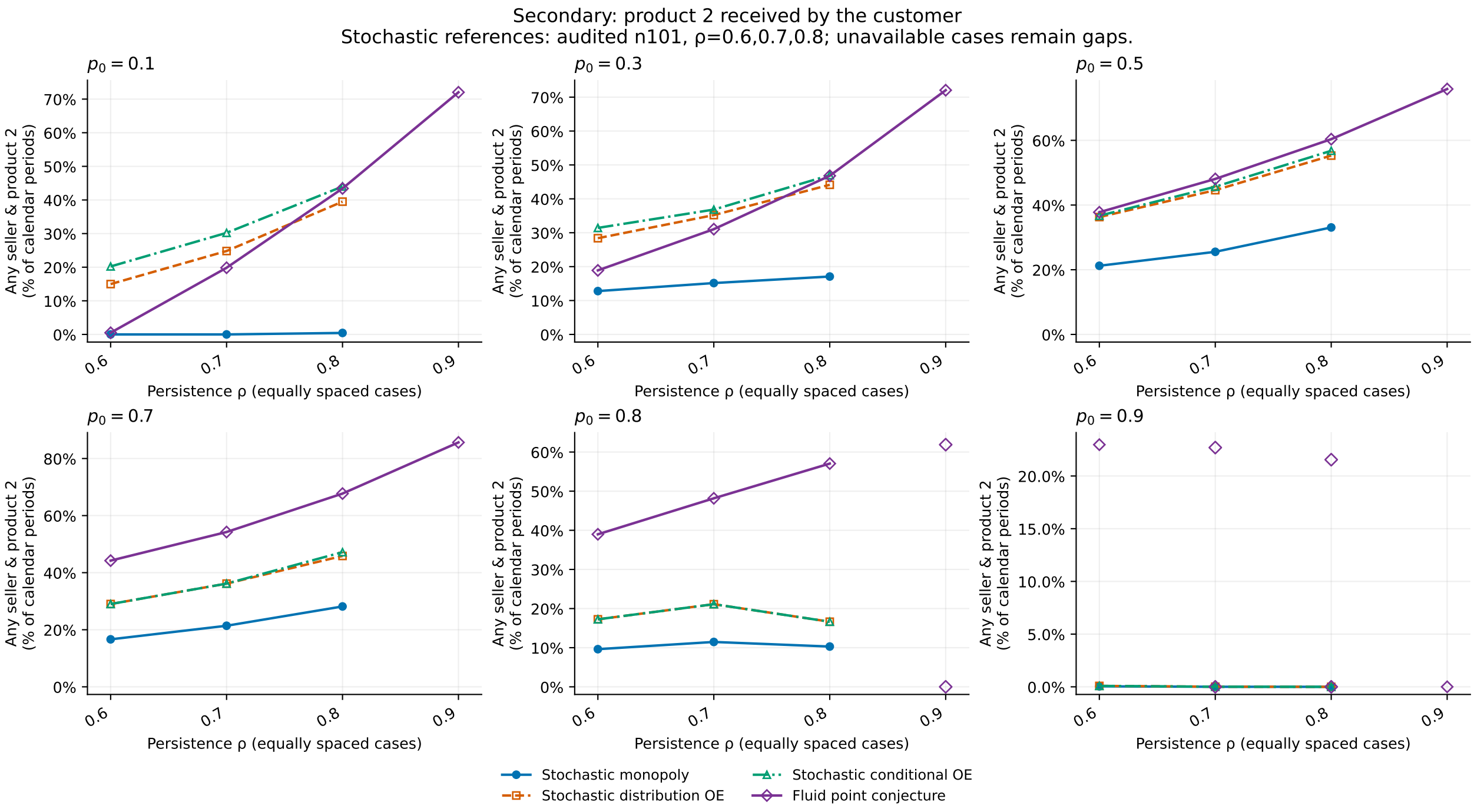
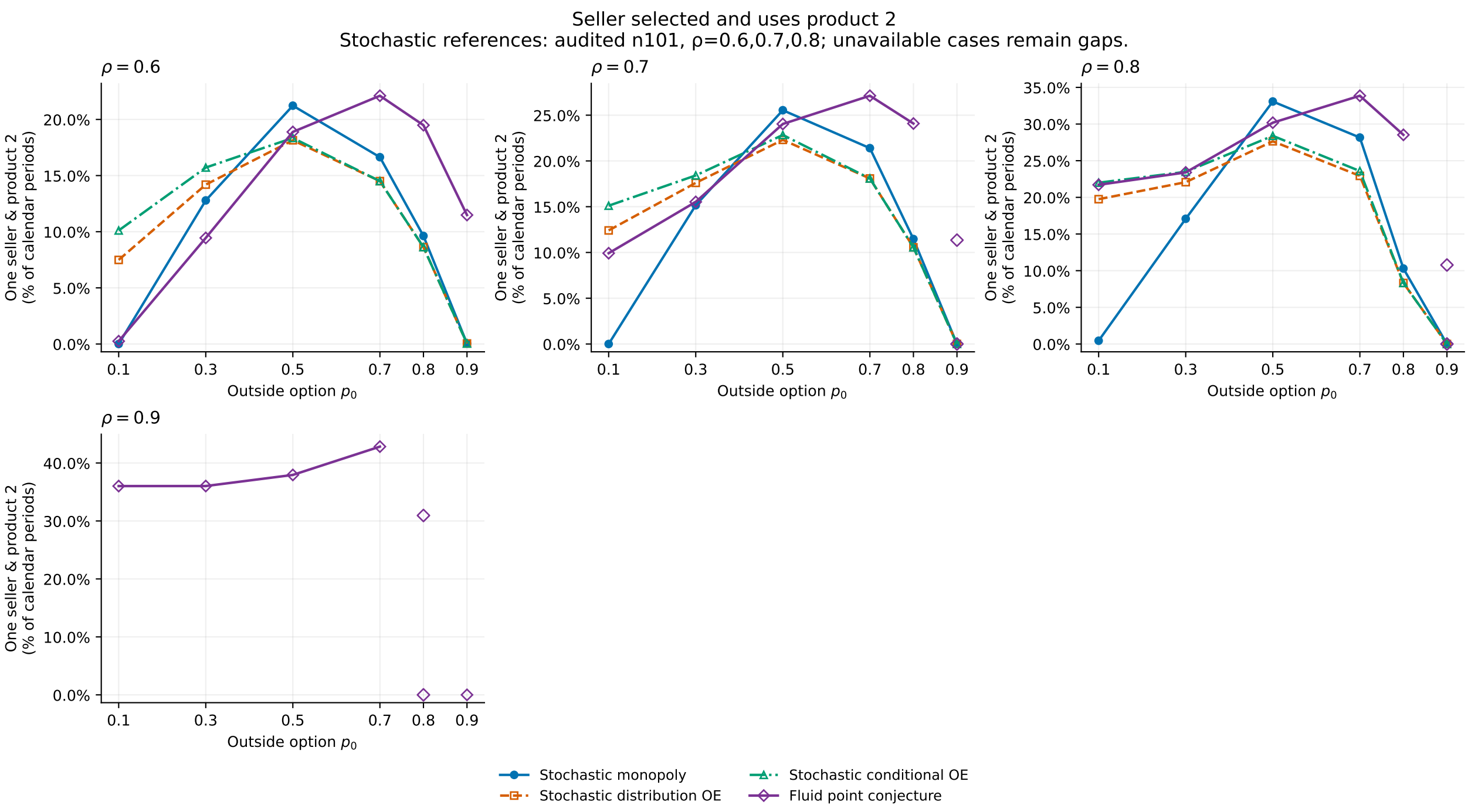
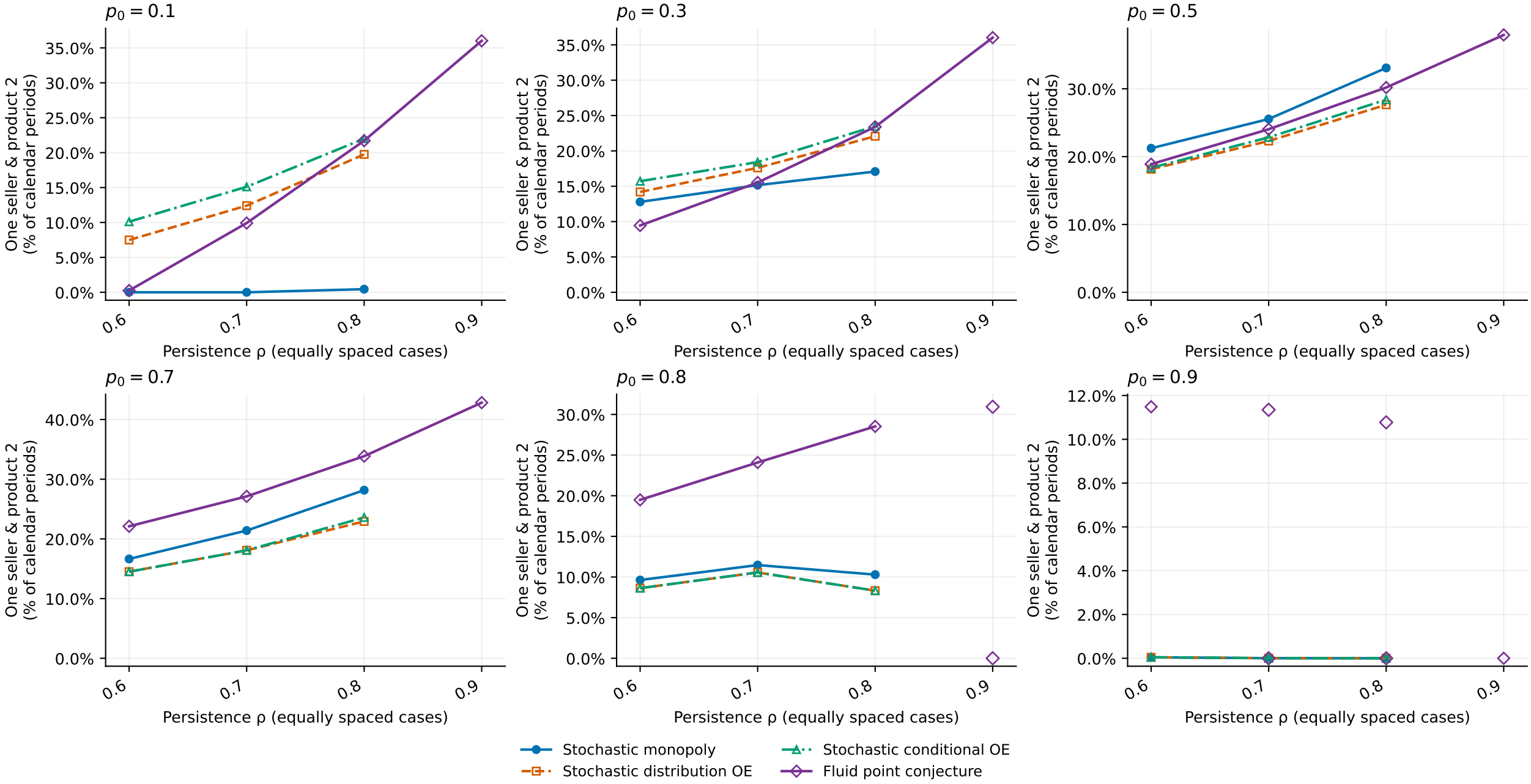

Primary: product 2 among own sales
Stochastic references: audited n101, ρ=0.6,0.7,0.8; unavailable cases remain gaps.
ρ = 0.6
ρ = 0.7
ρ = 0.8
ρ = 0.9
Product 2 (% of own sales)
100%
80%
60%
40%
20%
0%
0.1
0.3
0.5
0.7
0.8
0.9
Outside option p_0
Stochastic monopoly
Stochastic conditional OE
Stochastic distribution OE
Fluid point conjecture

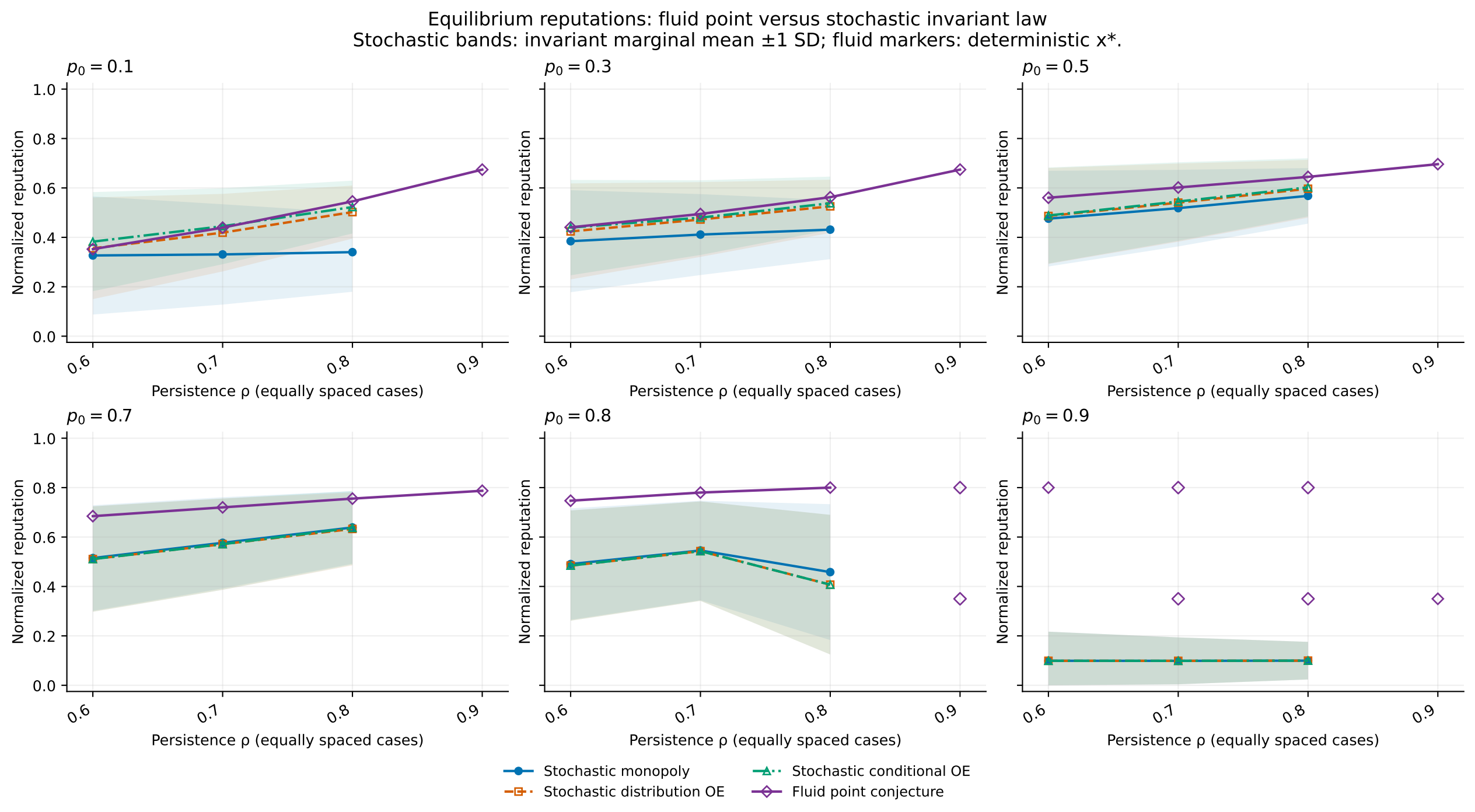

Primary: product 2 among own sales
Stochastic references: audited n101, ρ=0.6,0.7,0.8; unavailable cases remain gaps.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Product 2 (% of own sales)
100%
80%
60%
40%
20%
0%
0.6
0.7
0.8
0.9
Persistence ρ (equally spaced cases)
Stochastic monopoly
Stochastic conditional OE
Stochastic distribution OE
Fluid point conjecture

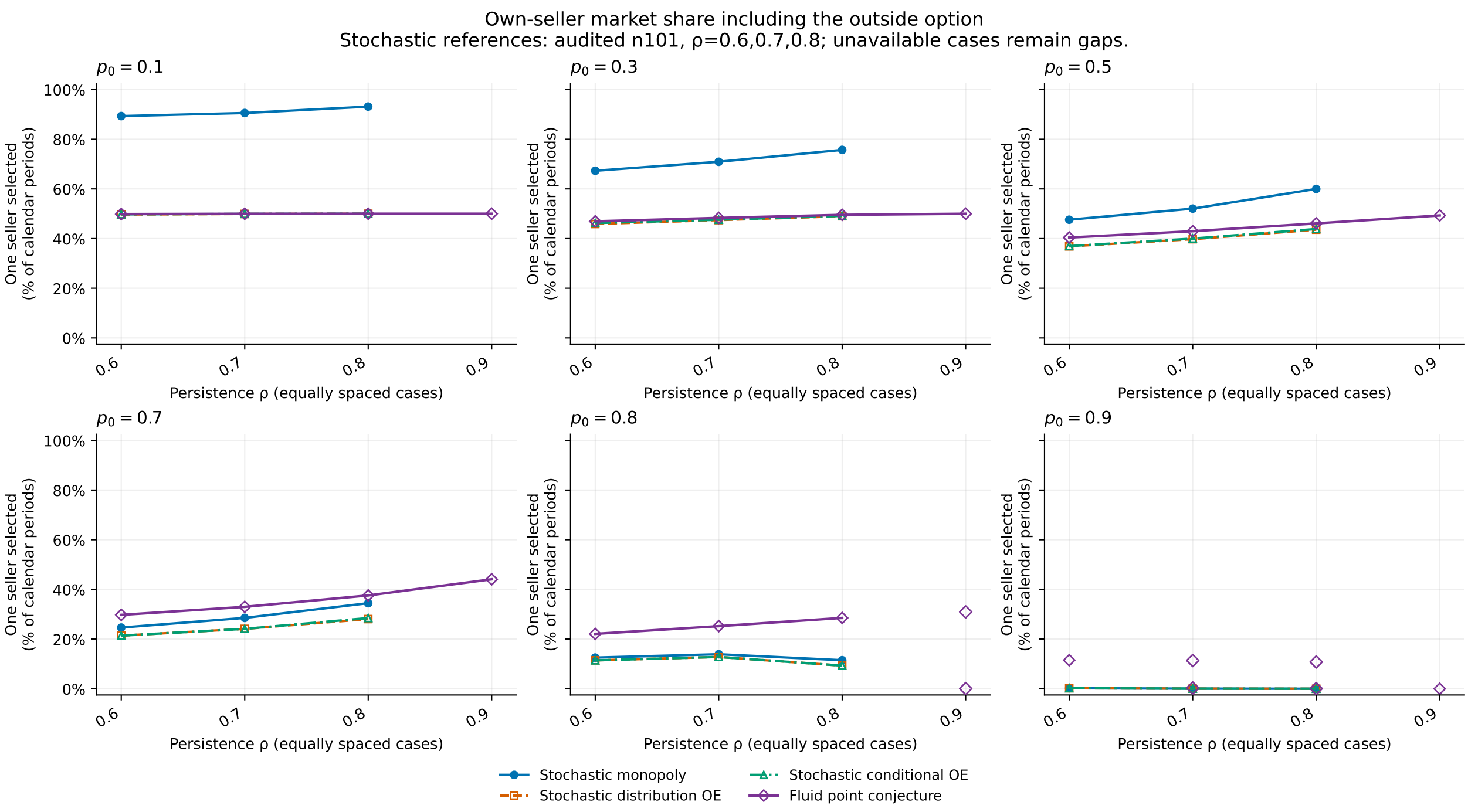

Secondary: product 2 received by the customer
Stochastic references: audited n101, ρ=0.6,0.7,0.8; unavailable cases remain gaps.
ρ = 0.6
ρ = 0.7
ρ = 0.8
ρ = 0.9
Any seller & product 2
(% of calendar periods)
Outside option p_0
0.1
0.3
0.5
0.7
0.8
0.9
Stochastic monopoly
Stochastic distribution OE
Stochastic conditional OE
Fluid point conjecture

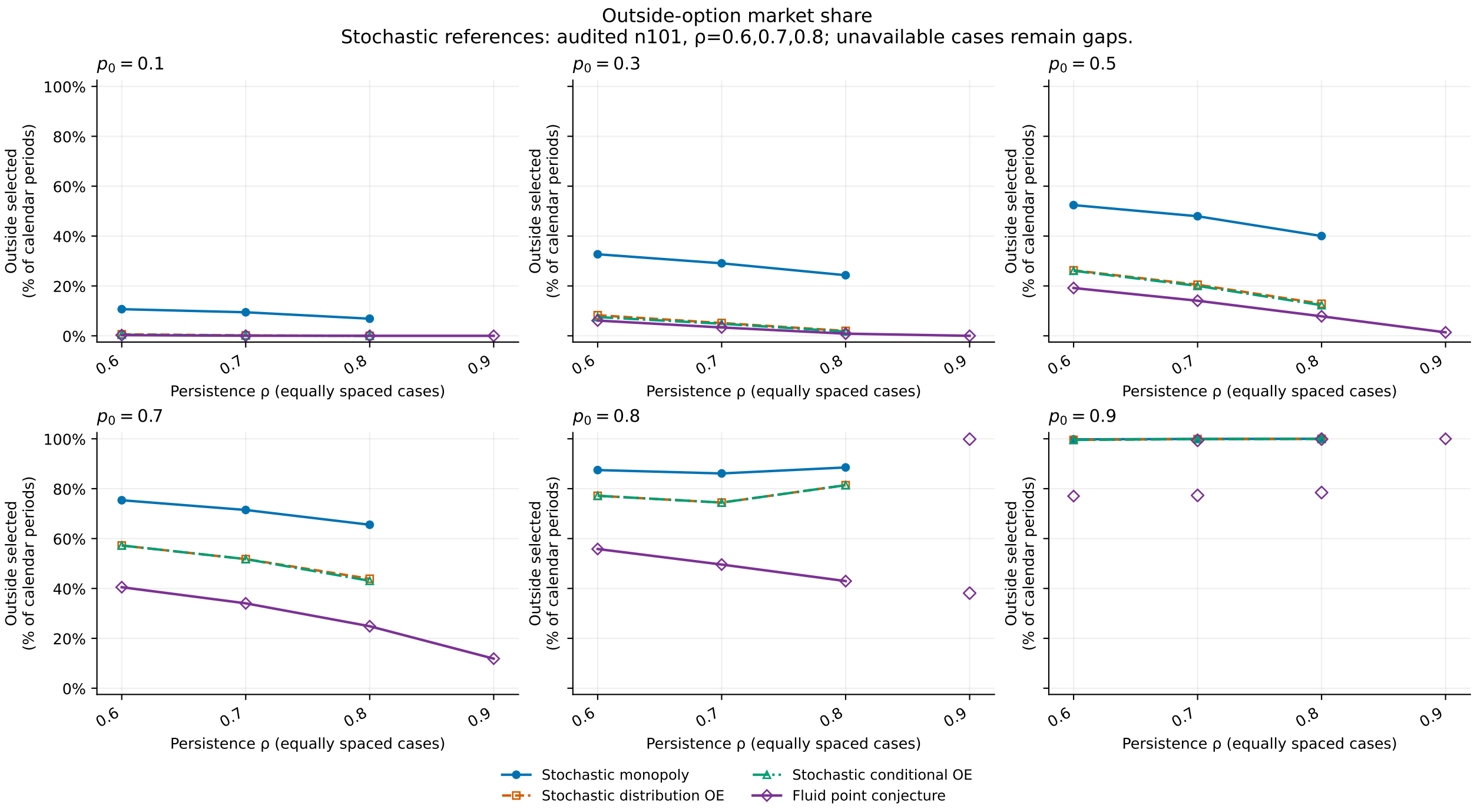

Secondary: product 2 received by the customer
Stochastic references: audited n101, ρ=0.6,0.7,0.8; unavailable cases remain gaps.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Any seller & product 2 (% of calendar periods)
Persistence ρ (equally spaced cases)
Stochastic monopoly
Stochastic distribution OE
Stochastic conditional OE
Fluid point conjecture

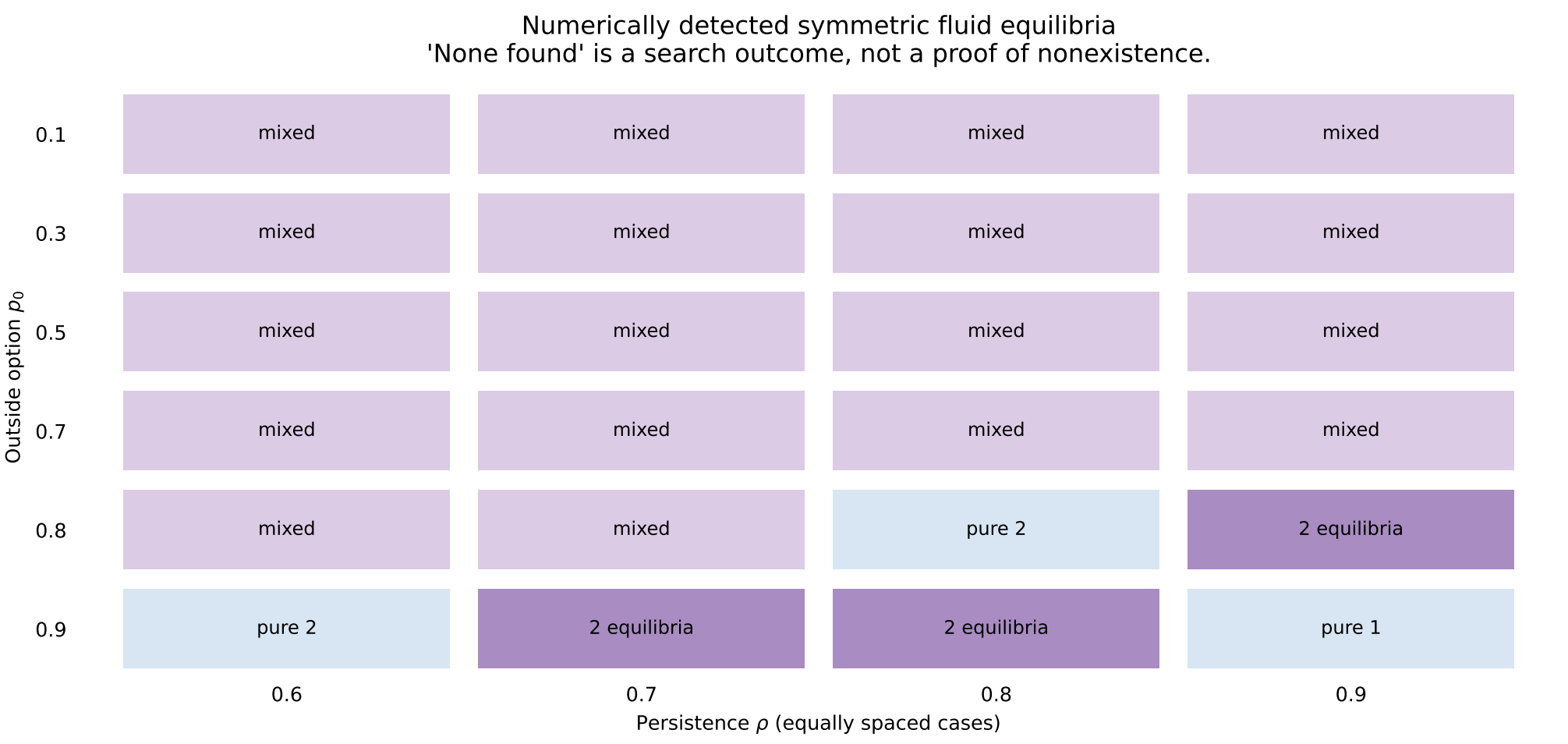

Seller selected and uses product 2
Stochastic references: audited n101, ρ=0.6,0.7,0.8; unavailable cases remain gaps.
ρ = 0.6
ρ = 0.7
ρ = 0.8
ρ = 0.9
One seller & product 2
(% of calendar periods)
Outside option p_0
0.0%
5.0%
10.0%
15.0%
20.0%
25.0%
30.0%
35.0%
40.0%
0.1
0.3
0.5
0.7
0.8
0.9
Stochastic monopoly
Stochastic distribution OE
Stochastic conditional OE
Fluid point conjecture

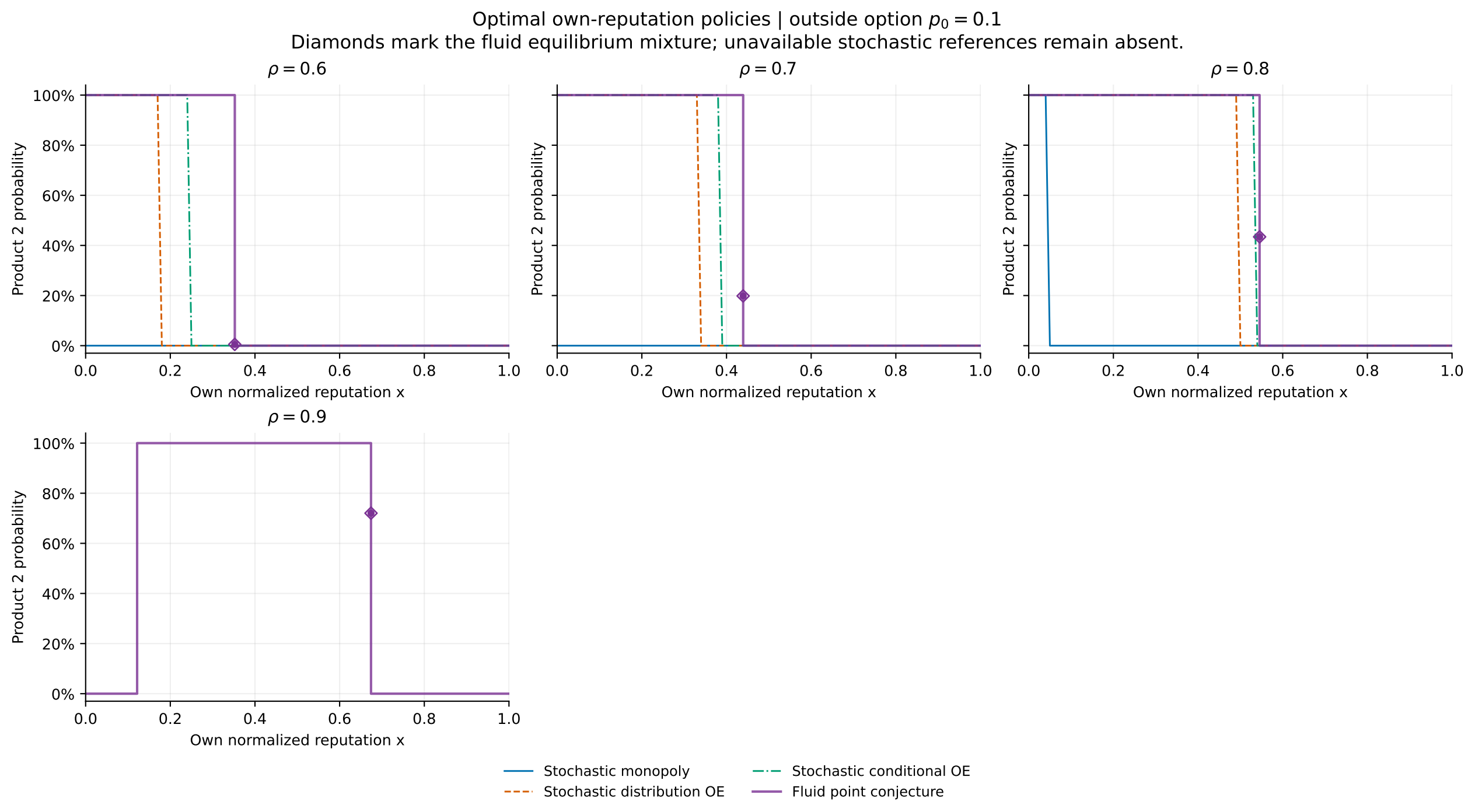

Seller selected and uses product 2
Stochastic references: audited n101, ρ=0.6,0.7,0.8; unavailable cases remain gaps.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
One seller & product 2 (% of calendar periods)
Persistence ρ (equally spaced cases)
Stochastic monopoly
Stochastic conditional OE
Stochastic distribution OE
Fluid point conjecture

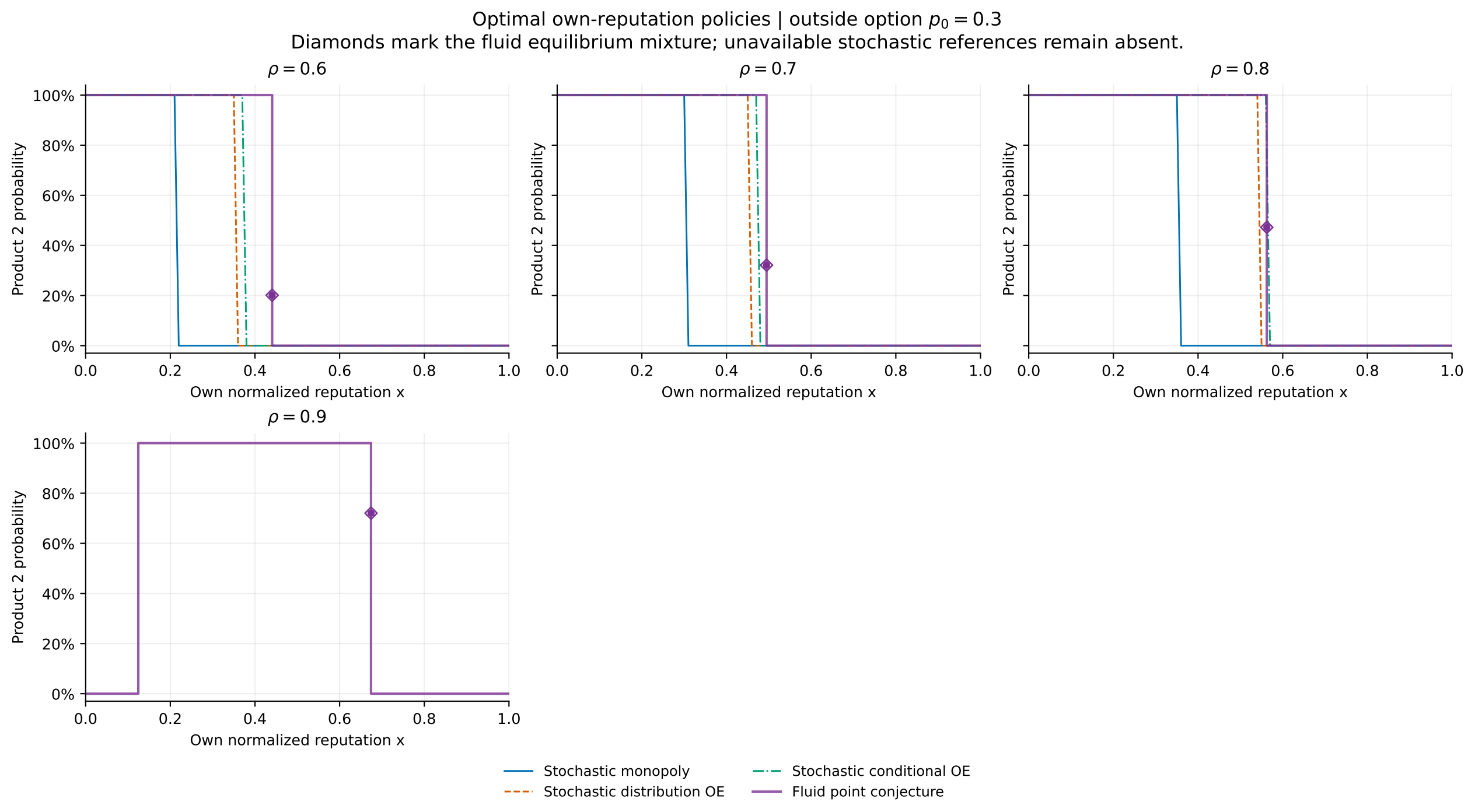

Equilibrium reputations: fluid point versus stochastic invariant law
Stochastic bands: invariant marginal mean ±1 SD; fluid markers: deterministic x*.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Normalized reputation
Persistence ρ (equally spaced cases)
0.0
0.2
0.4
0.6
0.8
1.0
0.6
0.7
0.8
0.9
Stochastic monopoly
Stochastic distribution OE
Stochastic conditional OE
Fluid point conjecture

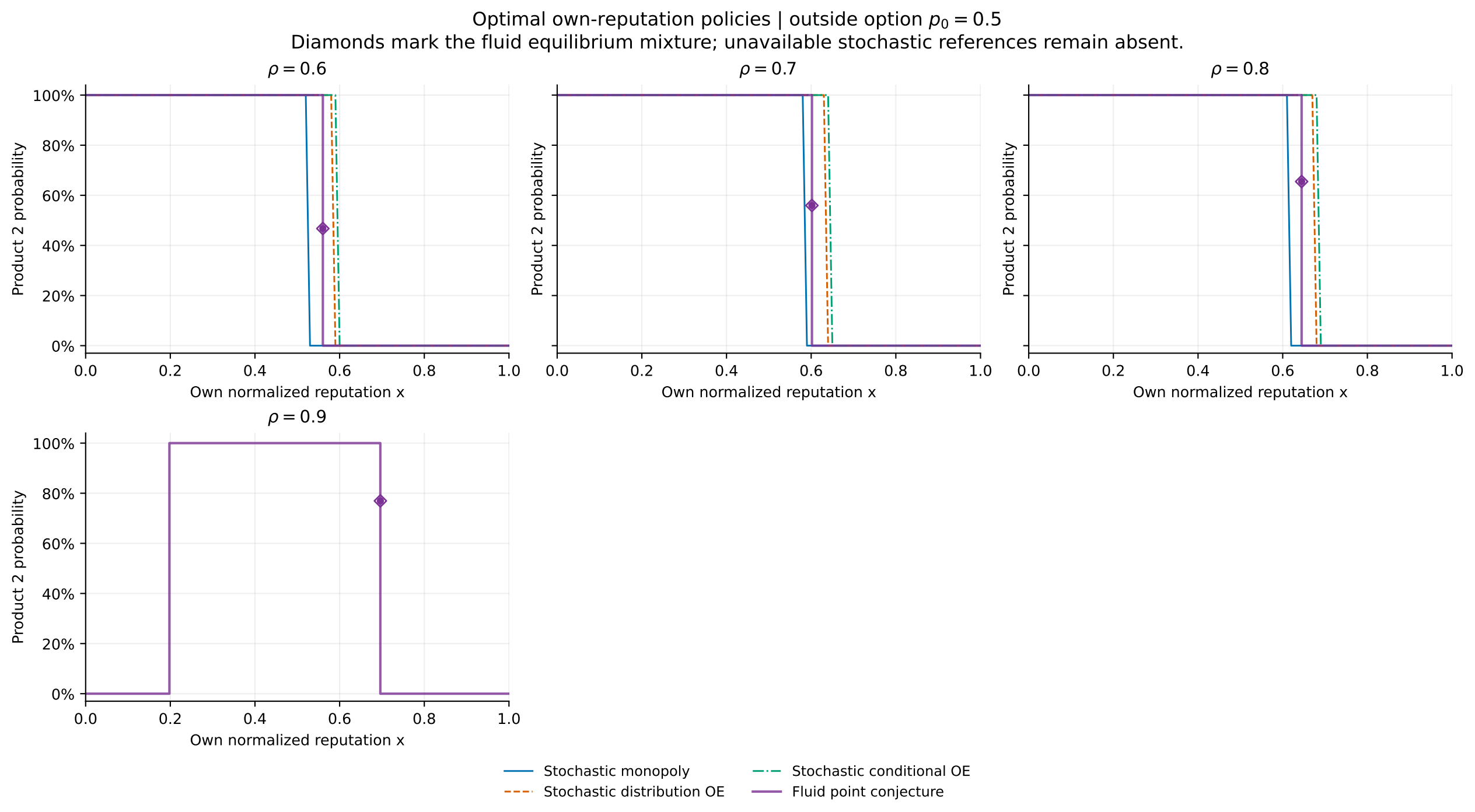

Own-seller market share including the outside option
Stochastic references: audited n101, ρ=0.6,0.7,0.8; unavailable cases remain gaps.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
One seller selected (% of calendar periods)
100%
80%
60%
40%
20%
0%
0.6
0.7
0.8
0.9
Persistence ρ (equally spaced cases)
Stochastic monopoly
Stochastic distribution OE
Stochastic conditional OE
Fluid point conjecture

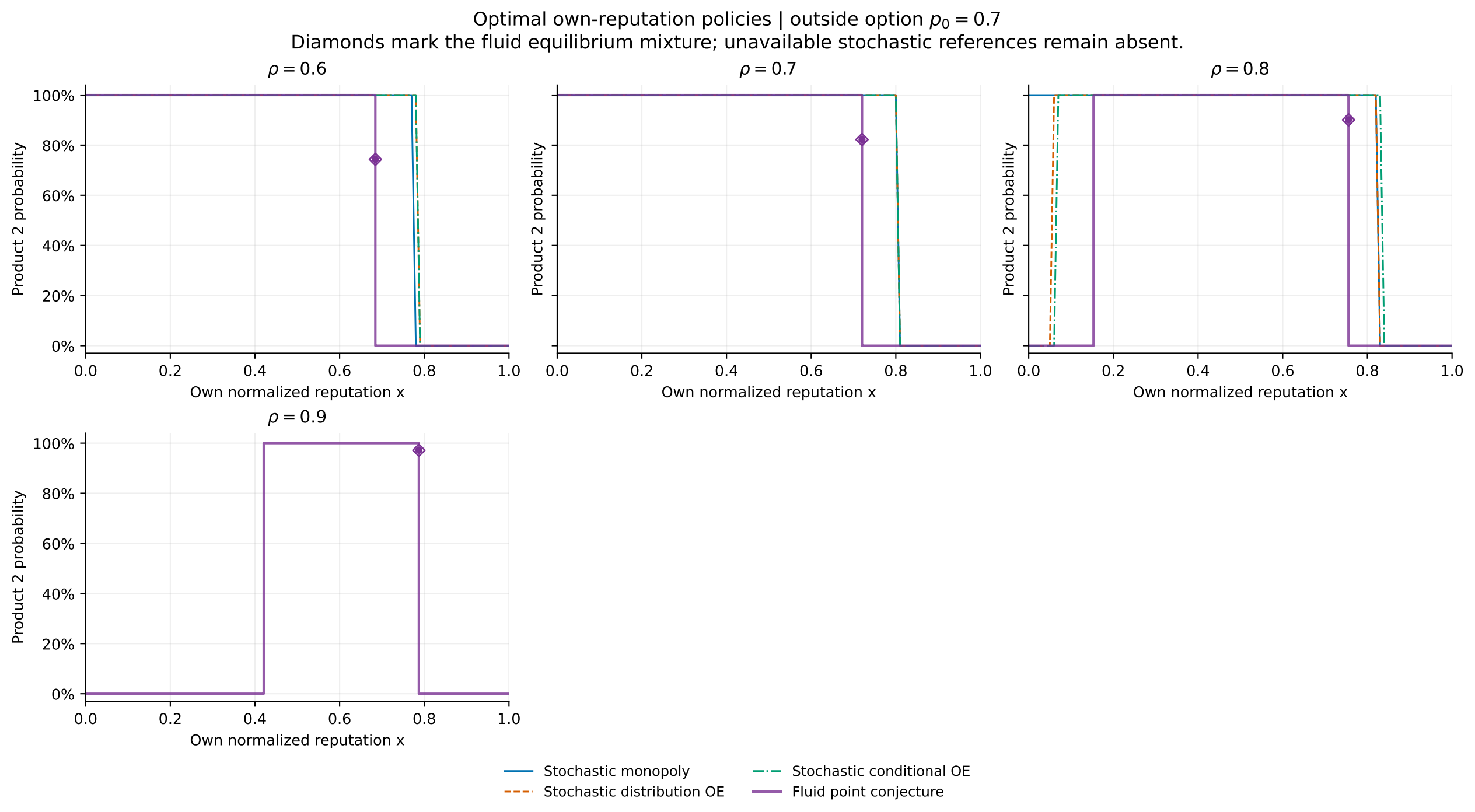

Outside-option market share
Stochastic references: audited n101, ρ=0.6,0.7,0.8; unavailable cases remain gaps.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Outside selected (% of calendar periods)
100%
80%
60%
40%
20%
0%
0.6
0.7
0.8
0.9
Persistence ρ (equally spaced cases)
Stochastic monopoly
Stochastic distribution OE
Stochastic conditional OE
Fluid point conjecture

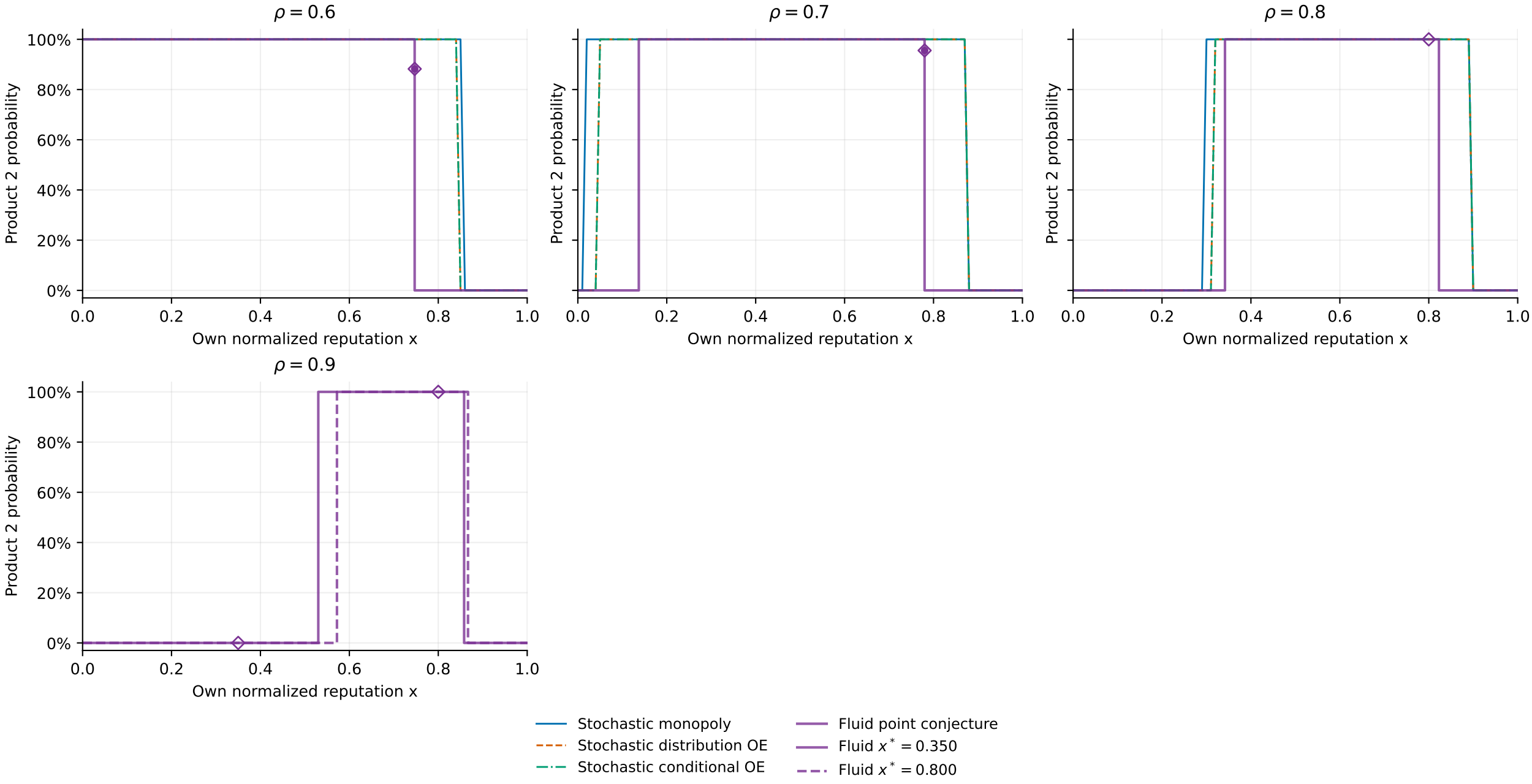

Numerically detected symmetric fluid equilibria
'None found' is a search outcome, not a proof of nonexistence.
Outside option p_0
0.1
0.3
0.5
0.7
0.8
0.9
mixed
pure 2
2 equilibria
pure 1
0.6
0.7
0.8
0.9
Persistence ρ (equally spaced cases)

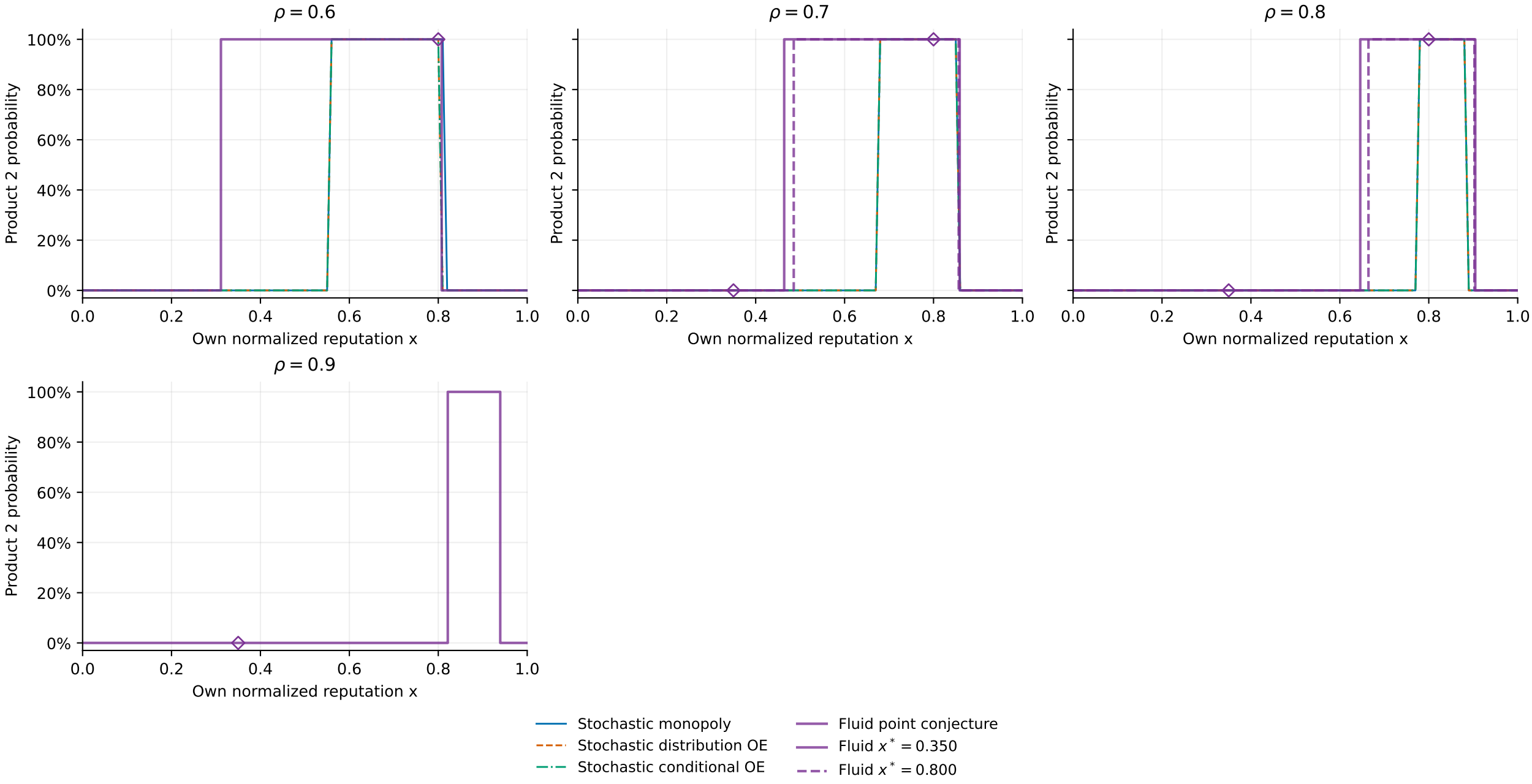

Optimal own-reputation policies | outside option $p_0 = 0.1$
Diamonds mark the fluid equilibrium mixture; unavailable stochastic references remain absent.
$\rho = 0.6$
$\rho = 0.7$
$\rho = 0.8$
$\rho = 0.9$
Product 2 probability
Own normalized reputation x
100%
80%
60%
40%
20%
0%
0.0
0.2
0.4
0.6
0.8
1.0
Stochastic monopoly
Stochastic distribution OE
Stochastic conditional OE
Fluid point conjecture

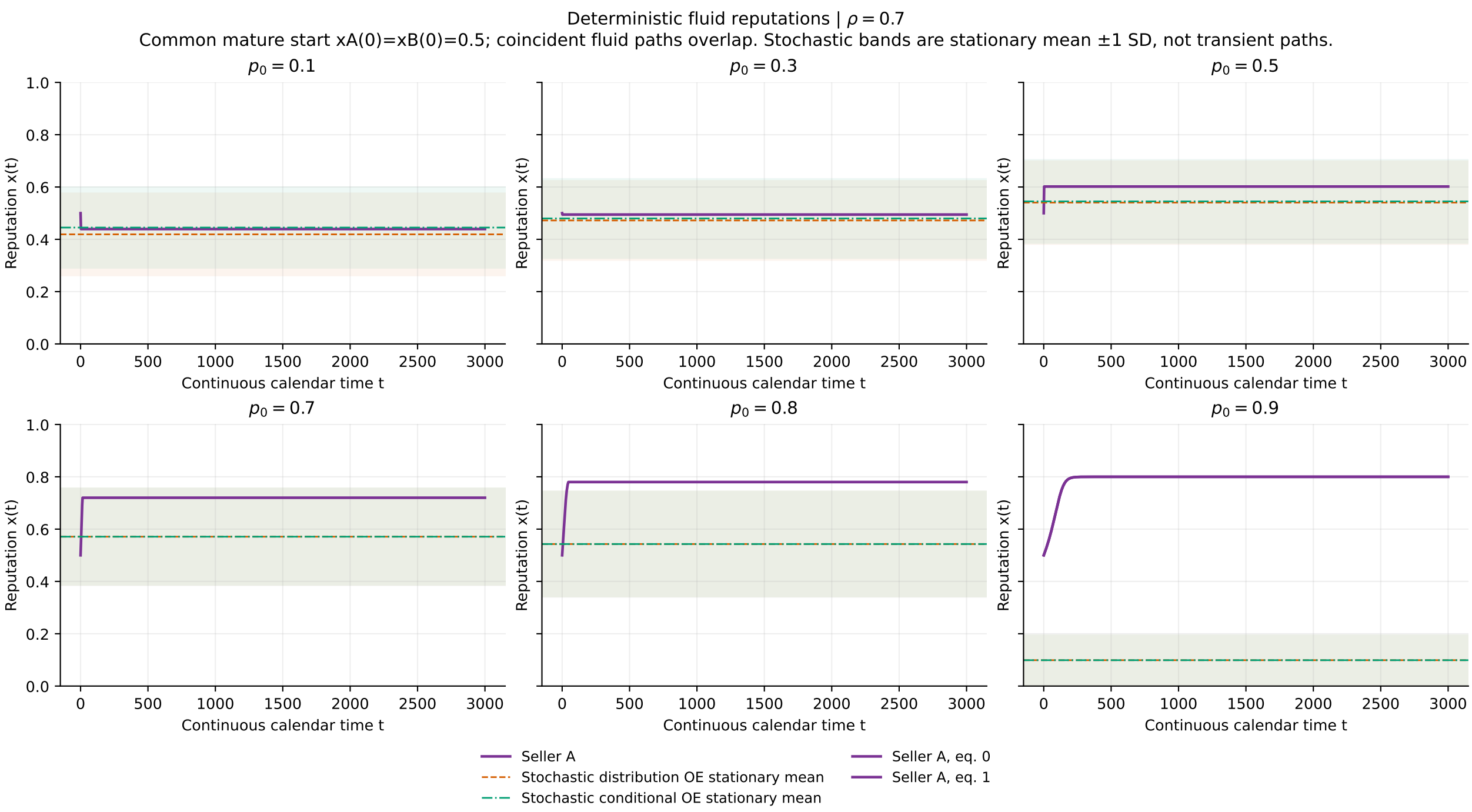

Optimal own-reputation policies | outside option $p_0 = 0.3$
Diamonds mark the fluid equilibrium mixture; unavailable stochastic references remain absent.
$\rho = 0.6$
$\rho = 0.7$
$\rho = 0.8$
$\rho = 0.9$
Product 2 probability
Own normalized reputation x
100%
80%
60%
40%
20%
0%
0.0
0.2
0.4
0.6
0.8
1.0
Stochastic monopoly
Stochastic distribution OE
Stochastic conditional OE
Fluid point conjecture

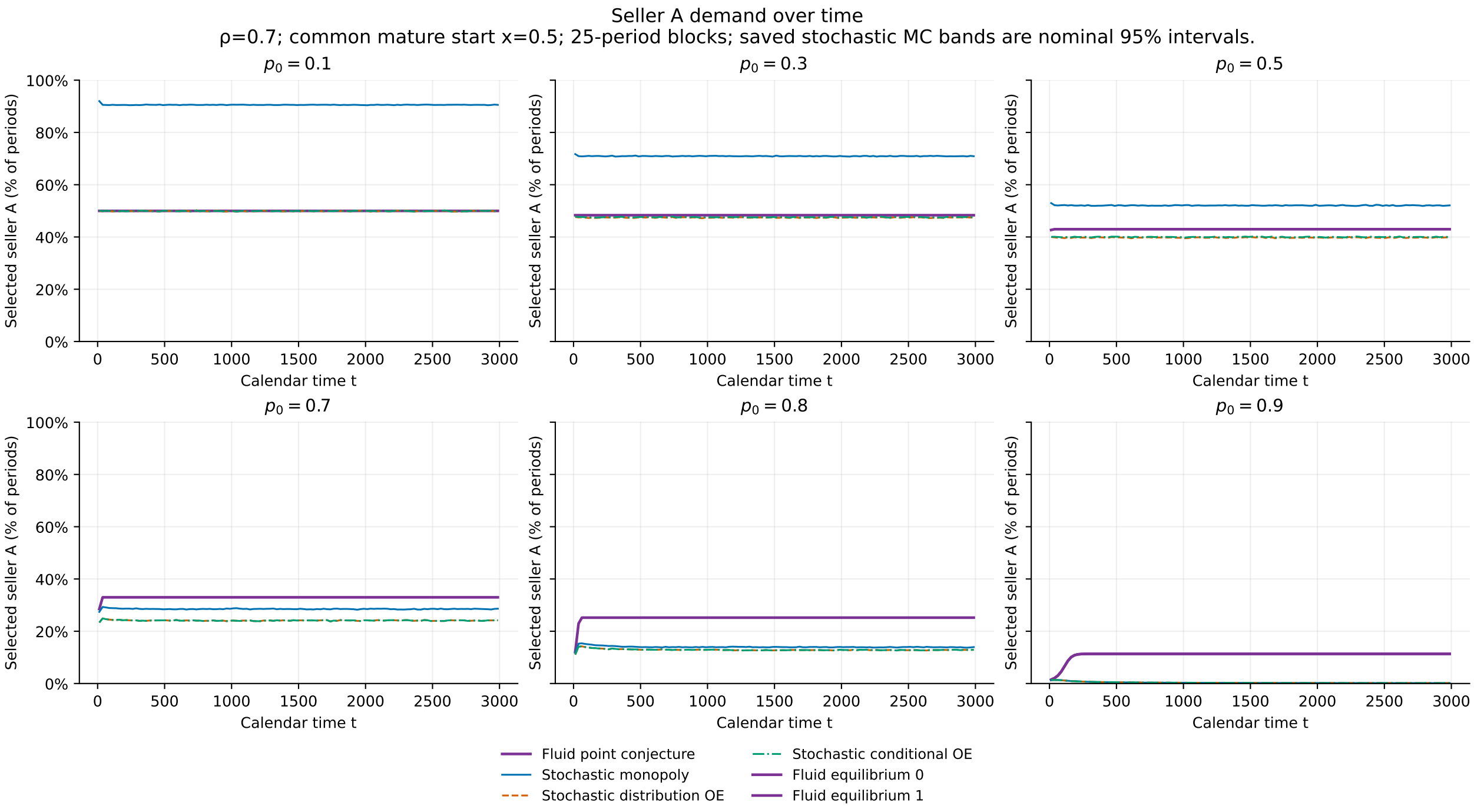

Optimal own-reputation policies | outside option $p_0 = 0.5$
Diamonds mark the fluid equilibrium mixture; unavailable stochastic references remain absent.
$\rho = 0.6$
$\rho = 0.7$
$\rho = 0.8$
$\rho = 0.9$
Product 2 probability
Own normalized reputation x
100%
80%
60%
40%
20%
0%
0.0
0.2
0.4
0.6
0.8
1.0
Stochastic monopoly
Stochastic distribution OE
Stochastic conditional OE
Fluid point conjecture

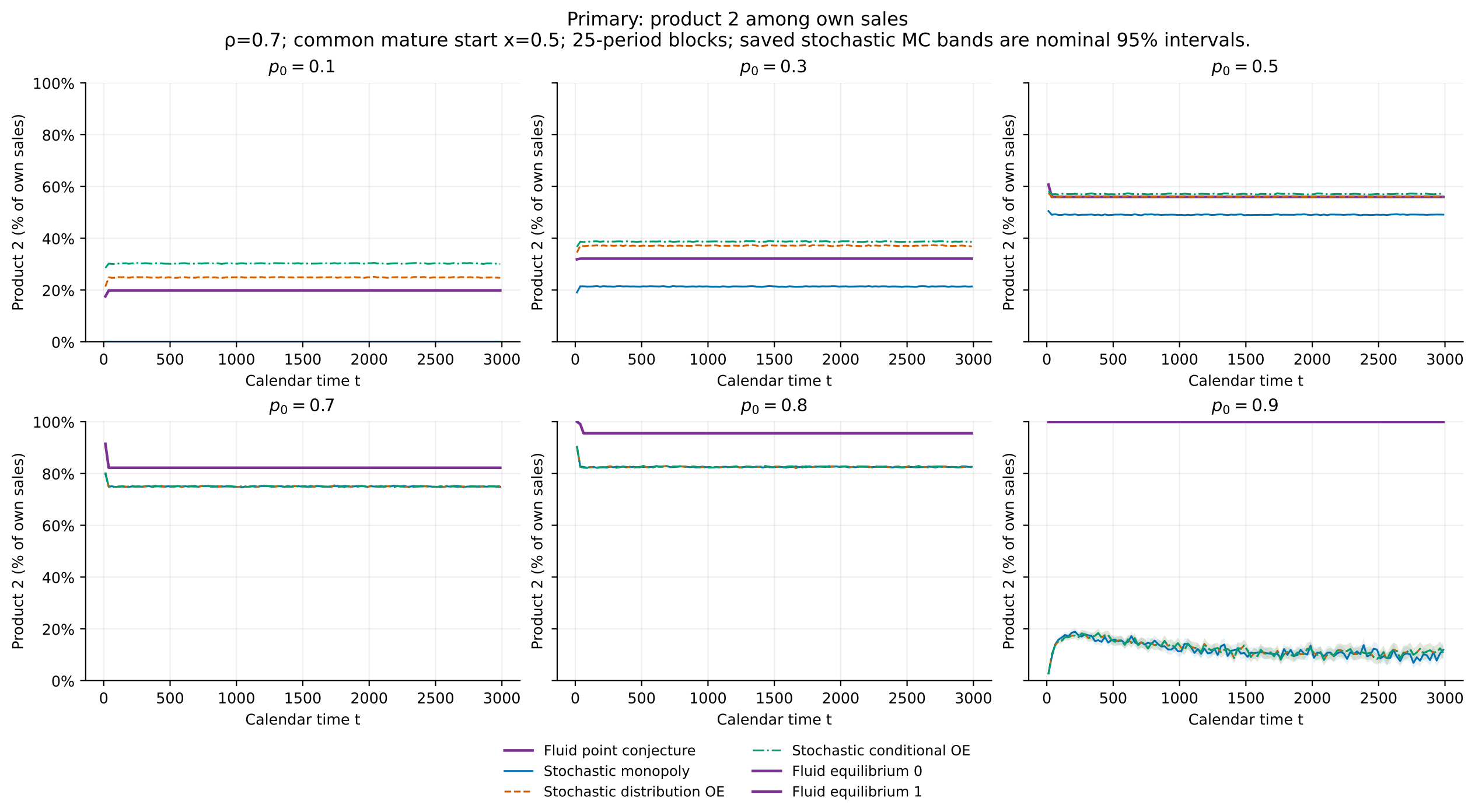

Optimal own-reputation policies | outside option $p_0 = 0.7$
Diamonds mark the fluid equilibrium mixture; unavailable stochastic references remain absent.
$\rho = 0.6$
$\rho = 0.7$
$\rho = 0.8$
$\rho = 0.9$
Product 2 probability
Own normalized reputation x
100%
80%
60%
40%
20%
0%
0.0
0.2
0.4
0.6
0.8
1.0
Stochastic monopoly
Stochastic distribution OE
Stochastic conditional OE
Fluid point conjecture

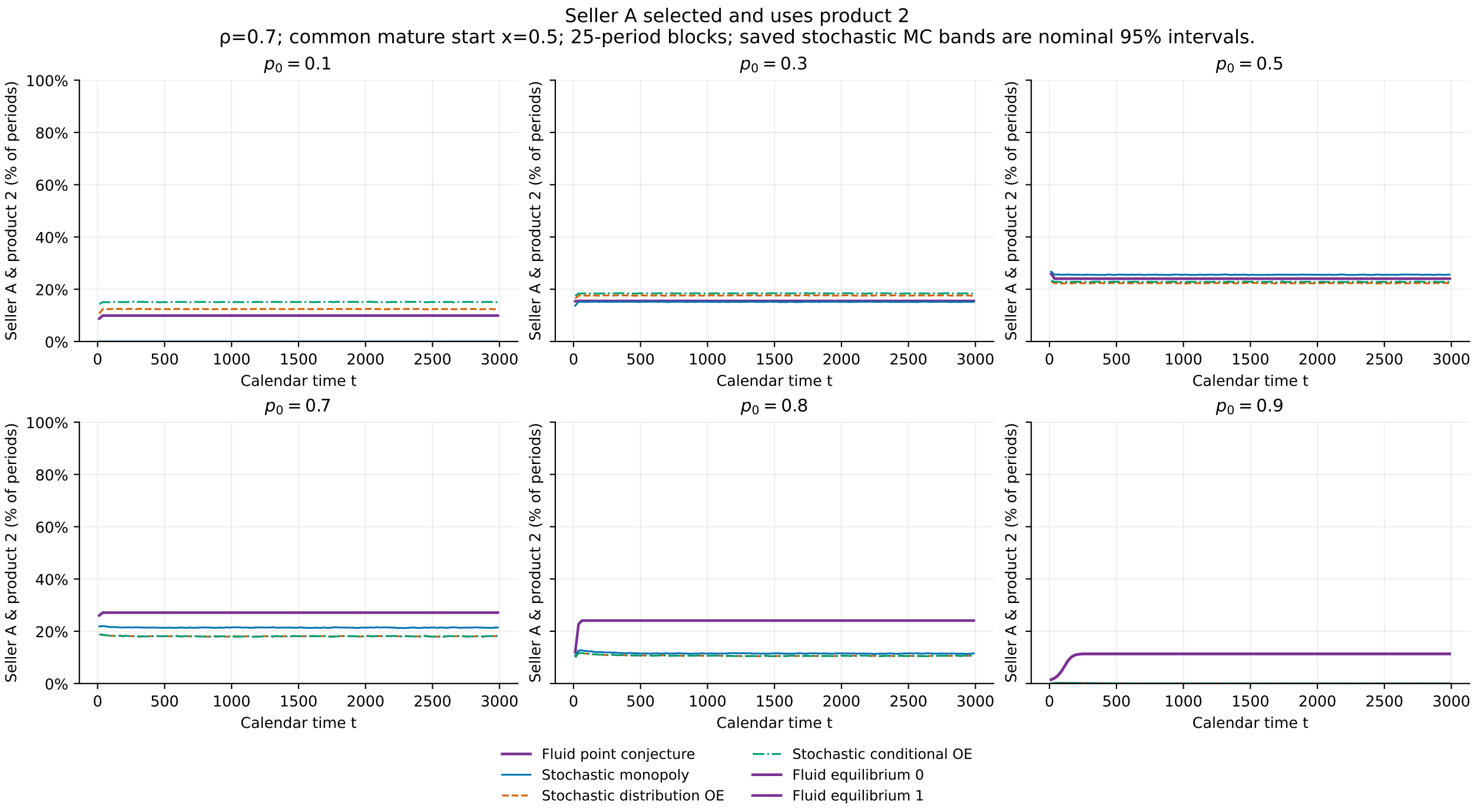

Optimal own-reputation policies | outside option $p_0 = 0.8$
Diamonds mark the fluid equilibrium mixture; unavailable stochastic references remain absent.
$\rho = 0.6$
$\rho = 0.7$
$\rho = 0.8$
$\rho = 0.9$
Product 2 probability
Own normalized reputation x
100%
80%
60%
40%
20%
0%
0.0
0.2
0.4
0.6
0.8
1.0
Stochastic monopoly
Stochastic distribution OE
Stochastic conditional OE
Fluid point conjecture
Fluid $x^* = 0.350$
Fluid $x^* = 0.800$

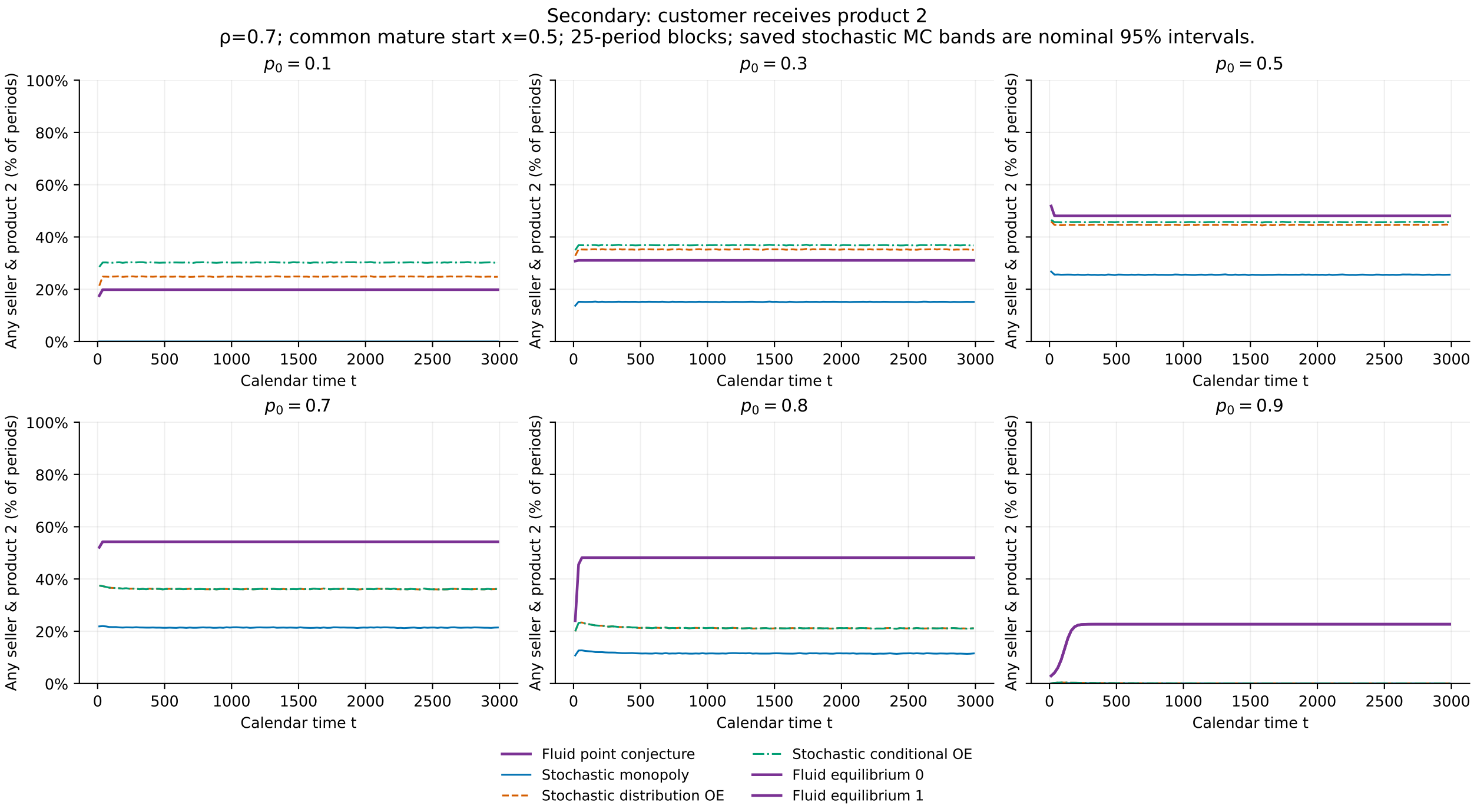

Optimal own-reputation policies | outside option $p_0 = 0.9$
Diamonds mark the fluid equilibrium mixture; unavailable stochastic references remain absent.
$\rho = 0.6$
$\rho = 0.7$
$\rho = 0.8$
$\rho = 0.9$
Product 2 probability
Own normalized reputation x
0%
20%
40%
60%
80%
100%
0.0
0.2
0.4
0.6
0.8
1.0
Stochastic monopoly
Stochastic distribution OE
Stochastic conditional OE
Fluid point conjecture
Fluid $x^* = 0.350$
Fluid $x^* = 0.800$


Deterministic fluid reputations | $\rho = 0.7$
Common mature start xA(0)=xB(0)=0.5; coincident fluid paths overlap. Stochastic bands are stationary mean ±1 SD, not transient paths.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Reputation x(t)
Continuous calendar time t
0.0
0.2
0.4
0.6
0.8
1.0
0
500
1000
1500
2000
2500
3000
Seller A
Stochastic distribution OE stationary mean
Stochastic conditional OE stationary mean
Seller A, eq. 0
Seller A, eq. 1


Seller A demand over time
ρ=0.7; common mature start x=0.5; 25-period blocks; saved stochastic MC bands are nominal 95% intervals.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Selected seller A (% of periods)
Calendar time t
0%
20%
40%
60%
80%
100%
0
500
1000
1500
2000
2500
3000
Fluid point conjecture
Stochastic monopoly
Stochastic distribution OE
Stochastic conditional OE
Fluid equilibrium 0
Fluid equilibrium 1


Primary: product 2 among own sales
ρ=0.7; common mature start x=0.5; 25-period blocks; saved stochastic MC bands are nominal 95% intervals.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Product 2 (% of own sales)
Calendar time t
0%
20%
40%
60%
80%
100%
0
500
1000
1500
2000
2500
3000
Fluid point conjecture
Stochastic monopoly
Stochastic distribution OE
Stochastic conditional OE
Fluid equilibrium 0
Fluid equilibrium 1


Seller A selected and uses product 2
ρ=0.7; common mature start x=0.5; 25-period blocks; saved stochastic MC bands are nominal 95% intervals.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Seller A & product 2 (% of periods)
Calendar time t
0%
20%
40%
60%
80%
100%
0
500
1000
1500
2000
2500
3000
Fluid point conjecture
Stochastic monopoly
Stochastic distribution OE
Stochastic conditional OE
Fluid equilibrium 0
Fluid equilibrium 1


Secondary: customer receives product 2
ρ=0.7; common mature start x=0.5; 25-period blocks; saved stochastic MC bands are nominal 95% intervals.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Any seller & product 2 (% of periods)
Calendar time t
0%
20%
40%
60%
80%
100%
0
500
1000
1500
2000
2500
3000
Fluid point conjecture
Stochastic monopoly
Stochastic distribution OE
Stochastic conditional OE
Fluid equilibrium 0
Fluid equilibrium 1